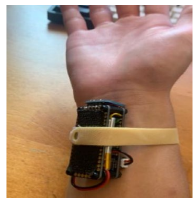
SC4172 Tiny Machine Machine Learning
Tutorial 3 – Machine Learning Case Study

Q1: Controlling 6DOF Robotic ARM via IMU Sensors

An IMU sensor board is used to control a robotic arm such that the robotic arm will emulate the movement of the human controller. IMU data are also used to train an NN model for gesture recognition.





IMU data sampling rate used is 100Hz. Initial testing shows the following accelerometer and gyro data excitation when performing the required gestures (Figure Q1-1). Assume the two graphs are representative of all gestures required.

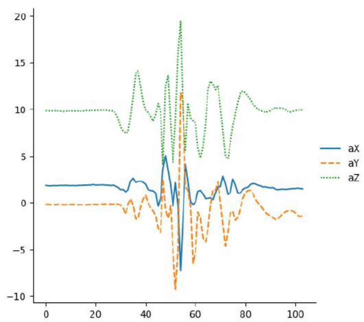


Figure 9 Accelerometer Waveform

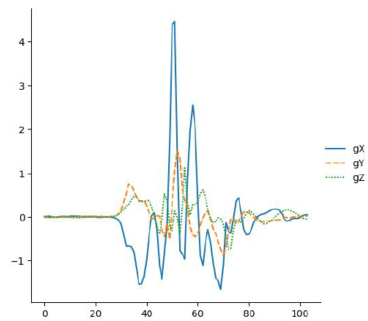


Figure 10 Gyroscope Waveform

Figure Q1-1: Accelerometer and Gyro Data

(i) What is a good window size to use for data capture?

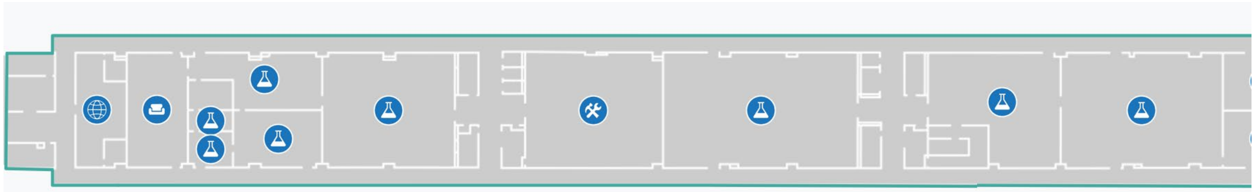
(ii) Does the dataset require any pre-processing to enable a more effective recognition model?

(iii) Data augmentation is often used to increase the size and randomness of the training dataset. For this system, in addition to adding random noise, one way is to offset the main pulse (excitation) in Figure Q1-1 horizontally. In doing so, part of the pulse may fall out of the capture frame. E.g. if the capture pulse is near the right of the frame, a positive offset would push part of the pulse beyond the right limit of the frame. Suggest one technique that can be applied to the augmented data to filter off data that does not have a complete pulse.

(iv) In actual deployment, the data capture may not be ideal, so the waveform may not be as illustrated in Figure Q1-1 but may have a horizontal offset to the left or right. One mitigating technique is to have sliding (and overlapping) windows within the capture frame to ensure essential data is fed into the NN model. Explain how this technique works. Suggest the duration of the capture frame, window size and time offset between windows. What are the trade-offs/factors to consider when deciding on these parameters?

(v) During NN model training, a typical way is to arrange the accelerometer and gyro data as a 1D tensor to feed into the NN model. Suggest other ways of arrangements and their advantages (if any).

Q2: Indoor asset location tracking via Wifi RSSI Fingerprint



Source:NTUmaps

GPS based location tracking typically does not work well indoors as GPS signal are often weak or block when indoors. A more reliable means is via detection of strength of wifi signals from Wifi Access Points (AP) in the vicinity of the object. The list of strength of wifi signals from wifi AP detected by the asset tag (wifi receiver) attached to the object forms a pattern (signature) which is unique to the object's location, e.g. [AP1=-43, AP2=-66, AP3=-33] may indicate that the object is at location (x_k, y_k). The continuously improving indoor location tracking performance of Apple and Google Maps is a result of training of data from GPS, Cellular and Wifi signals (plus other techniques). RSSI stands for Receiver Signal Strength Indicator, it is proportional to Strength of the RF Signals received.

(i) Common issues that developers may encounter during data collection are

a. Cannot differentiate between persistent and temporary Wifi signals. Temporary wifi signals may come from personal wireless routers, handphone hotspots etc.

b. No signals detected from certain Wifi APs.

What are the possible mitigating measures to counter these issues?

(ii) Typical data collected is a time series dataset of the RSSI of the surrounding APs. Simplest format of data arrangement (for training purpose) is (x_k, y_k) => [$AP1_T$, $AP2_T$, …APN_T, $AP1_{T+1}$, … APN_{T+1}, … APN_{T+M}], where N=AP number, T=time, M=time offset. Suggest other possible data formats and their pros/cons (if any).

(iii) Following from Q2(i)b above, if AP1 is located very far away and its signal is too weak to be detected at location (x_k, y_k), what RSSI value do we need to initialize $AP1_T$ to?

(iv) If Bluetooth-based Outpost is used instead of Wifi AP, and you have control over whether to design the asset tag (Bluetooth) to be a transmitter or receiver, which option would you choose and why? Discuss from perspective of power consumption. Asset tag is battery powered and Bluetooth Outpost is main powered.